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Paper 1: Integrating B Cell Differentiation Model With Real-World Data Informs Determinants

Bibliographic Information

- **Title:** Integrating B Cell Differentiation Model With Real-World Data Informs Determinants for Antibody-Secreting Cell Depletions in Systemic Sclerosis
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Paper Type: POPK_PKPD

Quick Take

This paper develops a quantitative systems pharmacology (QSP) model of B-cell differentiation to compare $CD19^+$ (inebilizumab) and $CD20^+$ (rituximab) depletion therapies in systemic sclerosis (SSc). By integrating real-world B-cell subset data from $n = 61$ untreated SSc patients, the model identifies the plasma cell-to-plasmablast (PC/PB) ratio as a key predictor of differential antibody-secreting cell (ASC) responses: high PC/PB ratio predicts poor rituximab response but stable inebilizumab depletion. The work provides a mechanistic framework for patient stratification and dosing optimization in B-cell-targeted therapies for SSc.

Executive Summary

The authors present a QSP model of B-cell differentiation and tissue distribution, calibrated with literature PK/PD data for inebilizumab and rituximab and validated against external rituximab studies. Real-world B-cell subset data from $n = 61$ untreated SSc patients were used to create virtual patients for clinical trial simulations. The model recapitulates uniform $CD19^+$ cell depletion by both drugs but reveals striking variability in ASC depletion with rituximab, driven by differences in baseline PC/PB ratio, drug-target binding affinity, and $CD20$ -rituximab elimination rate. Machine learning analysis confirmed PC/PB ratio as the most important baseline feature discriminating favorable ASC responses between the two therapies. This study demonstrates how integrating mechanistic modeling with real-world data can generate testable hypotheses for personalized B-cell depletion strategies in SSc.

Scientific Context & Motivation

Systemic sclerosis is a severe autoimmune disease with high mortality, where B cells and antibody-secreting cells (ASCs) play a central pathogenic role. $CD20$ -targeted therapy (rituximab) shows variable clinical responses and incomplete ASC depletion, while $CD19$ -targeted therapy (inebilizumab) depletes a broader B-cell lineage including ASCs. Prior clinical observations

suggested that baseline PC levels may predict inebilizumab response, but a comprehensive quantitative understanding of how B-cell subset composition influences differential drug effects was lacking. This study addresses that gap by building a mechanistic QSP model that integrates B-cell differentiation dynamics with drug PK/PD, enabling systematic exploration of patient-level factors driving ASC depletion variability.

✂ Methodological Snapshot

The authors developed a QSP model of B-cell differentiation (BCD) comprising compartments for total B cells, naïve B cells, memory B cells, plasmablasts, and plasma cells, with drug effects modeled via target-mediated depletion. PK models for inebilizumab (nonlinear TMDD with Michaelis-Menten approximation) and rituximab (linear clearance) were calibrated using literature data. Real-world B-cell subset data from $n = 61$ untreated SSc patients were used to create virtual patients by assigning individual baseline values. Global sensitivity analysis (Sobol method) identified key parameters, and machine learning (random forest with 5-fold cross-validation) was applied to simulated ASC depletion outcomes to identify baseline predictors. The model was validated against external rituximab studies.

☒ Structural Model Breakdown

The B-cell differentiation model consists of compartments representing different B-cell maturation stages: total B cells (TBs) differentiate into naïve B cells (NBs) and memory B cells (MBs), which further differentiate into plasmablasts (PBs) and then plasma cells (PCs). The model includes peripheral blood and tissue compartments. Drug effects: inebilizumab depletes $CD19^+$ cells (TBs, NBs, MBs, PBs, PCs) via target-mediated elimination; rituximab depletes $CD20^+$ cells (TBs, NBs, MBs, PBs) but not PCs (which are $CD20^-$). Methylprednisolone has a transient suppressive effect on B cells. The PK models for both drugs are two-compartment models with first-order absorption (IV administration). Inebilizumab includes a Michaelis-Menten term for TMDD. The full set of ODEs and parameters are provided in the supplementary materials (Table S4).

Detailed Methodological Analysis

Modeling Approach

Quantitative systems pharmacology (QSP) model implemented in MATLAB SimBiology. The B-cell differentiation model includes compartments for total B cells (TBs), naïve B cells (NBs), memory B cells (MBs), plasmablasts (PBs), and plasma cells (PCs) in peripheral blood and tissues. Drug effects are modeled as target-mediated depletion: inebilizumab binds $CD19$ on B cells and ASCs; rituximab binds $CD20$ on B cells (not on PCs). Methylprednisolone co-administration effects were also modeled. PK for inebilizumab used a two-compartment model with Michaelis-Menten approximation for TMDD; rituximab used a two-compartment linear model. Body weight was a covariate on clearance and tissue volumes.

Data Sources

Model calibration: literature data from inebilizumab phase 1 trial (0.1 – 10 mg/kg , $n = 4 - 7$ /dose) with $CD20^+$ and PC gene signature measurements; rituximab data from two SSc studies (375 mg/m^2 and 1000 mg regimens) with $CD19^+$, $CD19^+CD27^-$, and PB measurements. External validation: two additional rituximab studies with $CD20^+$ cell data. Real-world data: peripheral blood B-cell subsets (TBs, NBs, MBs, PBs, PCs) from $n = 61$ untreated SSc patients and $n = 51$ healthy controls from Keio University Hospital.

Estimation Methods

Nonlinear regression (weighted least squares) using MATLAB SimBiology’s parameter estimation functionality. Parameter precision reported as relative standard errors (RSE, %CV) from the variance-covariance matrix. Sequential estimation: first PK parameters, then MP effects, then drug-specific B-cell depletion parameters.

Model Evaluation

Goodness-of-fit plots comparing model predictions to observed data for PK and B-cell endpoints. External validation by simulating $CD20^+$ cell dynamics in two independent rituximab studies and comparing to observed ranges. Global sensitivity analysis (Sobol method) to identify influential parameters. Machine learning classifier performance assessed via 5-fold cross-validation (100 repeats) reporting accuracy and AUROC.

Covariate Analysis

Body weight included as a covariate on drug clearance and tissue volumes. Baseline B-cell subset ratios (PC/PB , $PC/CD20$, $PB/CD20$) and clinical features (autoantibody flags, IL-6 levels, mRSS, %FVC) were evaluated as predictors of differential ASC depletion using random forest and correlation analysis. PC/PB ratio was identified as the most important predictor.

Statistical Rigor Assessment

The study uses appropriate methods for parameter estimation (nonlinear regression with RSE) and sensitivity analysis (Sobol GSA). Machine learning analysis employs 5-fold cross-validation repeated 100 times, yielding high accuracy (0.982) and AUROC (0.986), though these metrics are on model-simulated outcomes rather than real clinical data. The small sample size ($n = 61$) limits the generalizability of the machine learning findings. No formal uncertainty propagation (e.g., Bayesian methods) was used for the QSP model predictions. Missing data were not an issue as RWD was cross-sectional. Sensitivity analyses were performed to identify influential parameters, but no formal model comparison (e.g., AIC/BIC) was reported for alternative structural models.

Key Findings

- 1) The QSP model accurately describes PK of inebilizumab (nonlinear TMDD) and rituximab (linear clearance) and captures B-cell subset dynamics in SSc.
- 2) Clinical trial simulations with $n = 61$ virtual patients show that both drugs achieve near-complete $CD19^+$ cell depletion, but ASC depletion is uniform with inebilizumab and highly variable with rituximab.
- 3) The

PC/PB ratio is the strongest baseline predictor of differential ASC response ($r = 0.909$, $p < 0.001$): patients with high PC/PB ratio have poor ASC depletion with rituximab but sustained depletion with inebilizumab. 4) Drug-specific parameters—inebilizumab’s $CD19$ depletion potency ($K_{C50, INEB}$) and rituximab’s $CD20$ binding affinity ($K_{ss, CD20-RTX}$) and elimination rate ($k_{int, CD20-RTX}$)—are major sensitivity factors for ASC responses. 5) The model was externally validated against two independent rituximab studies, accurately predicting $CD20^+$ cell depletion and repopulation.

🔗 Clinical & Regulatory Implications

The model suggests that patients with a high PC/PB ratio (i.e., more long-lived plasma cells relative to short-lived plasmablasts) are likely to have inadequate ASC depletion with rituximab and may benefit more from inebilizumab. This provides a rationale for baseline B-cell subset phenotyping to guide treatment selection in SSc. The model also highlights that drug-target binding affinity and elimination kinetics (especially for rituximab via FcγR-mediated clearance) are important determinants of response, suggesting that dosing adjustments or alternative anti- $CD20$ antibodies with different binding properties could improve outcomes. From a regulatory perspective, the model could support dose optimization and patient enrichment strategies for phase 2/3 trials of B-cell-depleting therapies in SSc.

Strengths & Limitations

Strengths

- Mechanistic QSP model integrating B-cell differentiation, tissue distribution, and drug PK/PD for two monoclonal antibodies
- Integration of real-world B-cell subset data from $n = 61$ untreated SSc patients to create realistic virtual populations
- Rigorous global sensitivity analysis and machine learning (random forest with cross-validation) to identify key predictors
- External validation of the model against independent rituximab studies
- Clinically actionable finding: PC/PB ratio as a potential biomarker for treatment selection

Limitations (Acknowledged by Authors)

- Small sample size ($n = 61$) for RWD, with relatively mild SSc severity
- Model does not link ASC depletion to clinical outcomes (mRSS, FVC) or autoantibody levels
- Tissue dynamics of ASCs in secondary lymphoid organs and bone marrow not validated
- FCGR3A polymorphism effects not explicitly incorporated into the model
- Limited longitudinal post-treatment data in RWD; baseline associations are model-derived

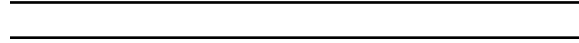
Limitations (Expert Review)

- PK model for rituximab used linear clearance; TMDD may be relevant at lower doses or in different populations

- Model calibration relied on median data from small clinical trials ($n = 4 - 7$ per dose group), increasing uncertainty
- No formal uncertainty quantification (e.g., Bayesian methods) for parameter estimates beyond RSE
- Machine learning analysis performed on model-simulated outcomes, not on actual clinical endpoints
- Potential overfitting in random forest despite cross-validation due to small number of virtual patients

Generalizability

The B-cell biology captured in the model is applicable across SSc severities, but the RWD cohort had relatively mild disease (mean mRSS ~10, mean %FVC ~95%). The model's predictions for ASC depletion may not directly translate to patients with severe organ involvement or those on concomitant immunosuppressants. The framework is generalizable to other autoimmune diseases where B-cell depletion is used, but drug-specific parameters would need recalibration.



Figures & Tables

- **Figure 1:** Overview of the clinical trial simulation workflow: integration of B-cell differentiation model with RWD from SSc patients to simulate $CD19^+$ and ASC dynamics under inebilizumab and rituximab.
 - *Significance:* Provides a visual summary of the modeling framework and how RWD feeds into virtual patient generation and simulation.
- **Figure 2:** Goodness-of-fit plots for inebilizumab PK (A), $CD20^+$ cells (B), PC gene signatures (C), and rituximab PK (D), $CD19^+$ cells (E), and PBs (F).
 - *Significance:* Demonstrates model calibration performance against observed clinical data for both drugs and multiple B-cell endpoints.
- **Figure 3:** Distribution of B-cell subsets (TBs, NBs, MBs, PBs, PCs) in SSc patients vs. healthy controls (A), histograms showing lognormal distributions (B), and individual patient profiles (C).
 - *Significance:* Characterizes the RWD used for model development and highlights patient heterogeneity, especially the PC/PB ratio.
- **Figure 4:** Simulated time courses of $CD19^+$ cells (A, C) and ASCs (B, D) for inebilizumab (top) and rituximab (bottom) in $n = 61$ virtual patients.
 - *Significance:* Key result showing uniform $CD19$ depletion but highly variable ASC depletion with rituximab, motivating the search for predictors.
- **Figure 5:** Proportion of VPs achieving ASC suppression thresholds (A), duration of $\geq 50\%$ ASC suppression per patient (B), and random forest feature importance (C) identifying PC/PB ratio as top predictor.
 - *Significance:* Central figure establishing PC/PB ratio as the most important baseline factor differentiating ASC responses between the two drugs.

- **Table 1:** Summary of clinical data used for model calibration and validation, including dose regimens, measurements, and references for inebilizumab, methylprednisolone, and rituximab studies.
 - *Significance:* Documents the data sources underpinning the model, essential for reproducibility and assessing data quality.
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Code & Reproducibility Assessment

The model was developed using MATLAB SimBiology. No public code repository or model files were provided. The paper describes parameter estimation methods and provides parameter tables (Table S4), but full model equations and implementation details are not openly available.

Supplementary Materials

Supplementary materials include Table S2 (demographics of RWD cohort), Table S3 (parameter ranges for GSA), Table S4 (estimated parameter values with RSE), Figure S3 (modeling workflow), Figure S4 (GSA results and external validation), Figure S5 (simulation design), Figure S6 (individual patient ASC trajectories), and Figure S7 (additional correlation analyses and GSA for ASCs). These provide critical details on parameter estimates, sensitivity analysis, and validation results.

Future Directions

- 1) Extend the model to incorporate tissue compartments (lymph nodes, bone marrow) to better understand ASC dynamics and their link to clinical outcomes.
 - 2) Integrate FCGR3A polymorphism effects on rituximab-mediated depletion to refine patient stratification.
 - 3) Link ASC depletion to autoantibody levels and clinical endpoints (mRSS, FVC) using longitudinal data from ongoing trials.
 - 4) Apply the framework to other autoimmune diseases (e.g., SLE, NMOSD) where both *CD19* and *CD20* therapies are used.
 - 5) Perform prospective validation of the *PC/PB* ratio as a predictive biomarker in a clinical trial.
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Expert Commentary

This study represents a thoughtful application of QSP to a clinically relevant question in autoimmune disease. The integration of real-world B-cell subset data is a strength, as it grounds the model in actual patient heterogeneity. The identification of *PC/PB* ratio as a predictor is biologically plausible: rituximab spares *CD20*-negative PCs, so patients with a high *PC*-to-*PB* ratio have a larger pool of long-lived PCs that are resistant to *CD20* depletion. The machine learning analysis adds rigor but should be interpreted cautiously given the small sample. From a regulatory perspective, this model could support dose optimization and patient selection for phase 2/3 trials. A key next step would be to validate the *PC/PB* ratio threshold prospectively. The authors' decision to use linear PK for rituximab is pragmatic given data limitations, but future work should explore TMDD when richer data become available. Overall, this is a well-executed study that advances the field of QSP in immunology.

Bottom Line

This QSP model provides a mechanistic understanding of why *CD19*-targeted therapy (inebilizumab) achieves more consistent ASC depletion than *CD20*-targeted therapy (rituximab) in SSc, with the baseline *PC/PB* ratio emerging as a key determinant. For practicing pharmacometricians, this work exemplifies how integrating real-world patient data with mechanistic modeling can identify clinically relevant biomarkers and support personalized treatment decisions. The model framework is extensible to other B-cell-targeted therapies and could be linked to downstream clinical outcomes once appropriate data become available.

☒ Figures

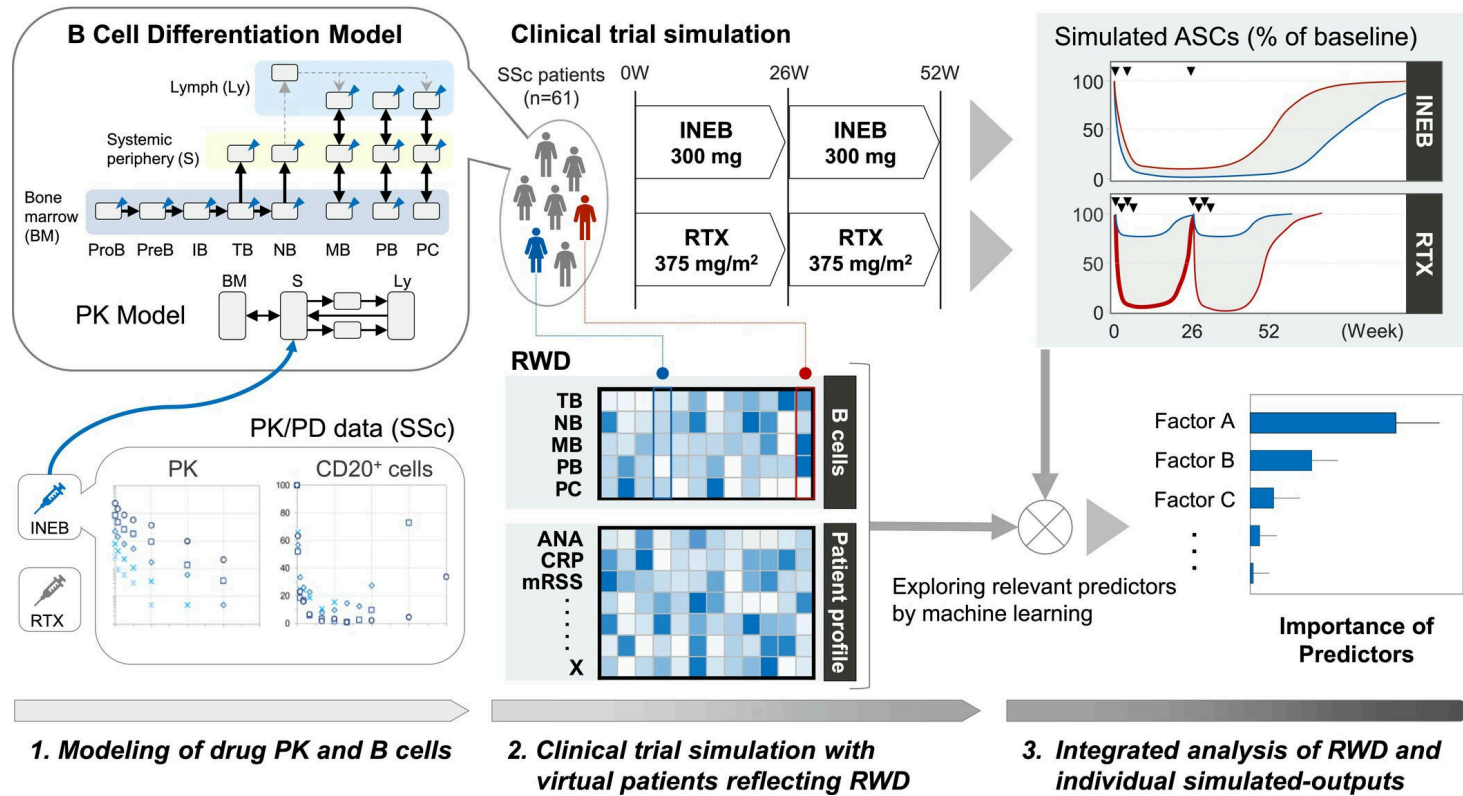


Figure 1: Overview of clinical trial simulation using B cell differentiation model and integrated analysis with RWD from patients with SSc. ASCs, antibody-secreting cell;

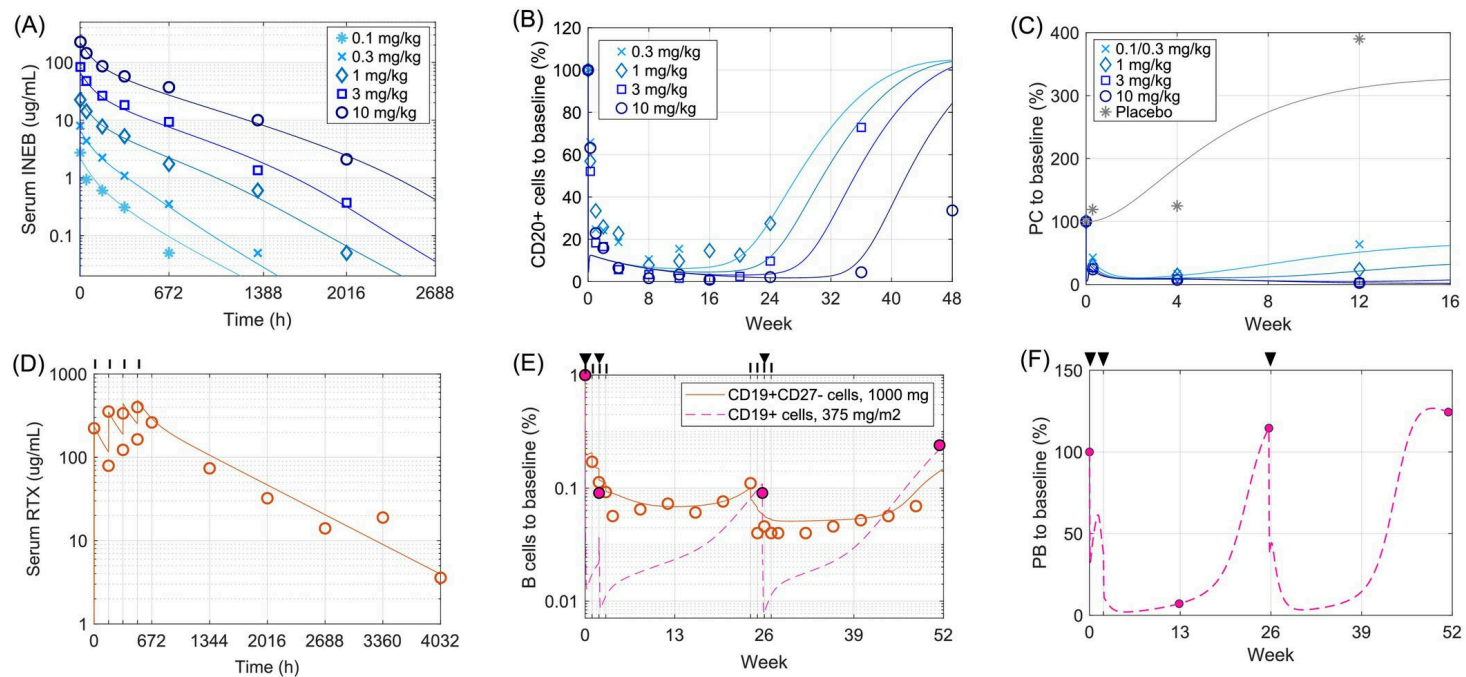


Figure 2: Goodness-of-fit plots of fitted time-courses of drug PK and B cell subsets with the BCD model. For inebilizumab (INEB), its serum concentration–time profiles (A)

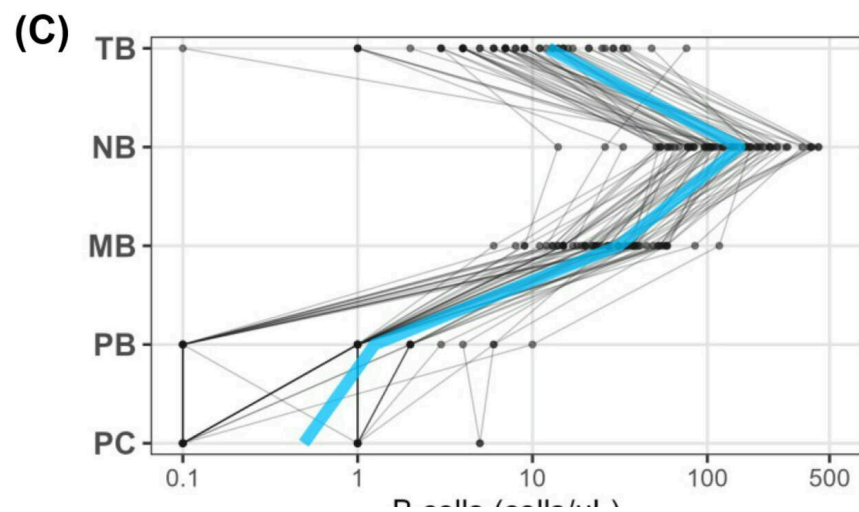
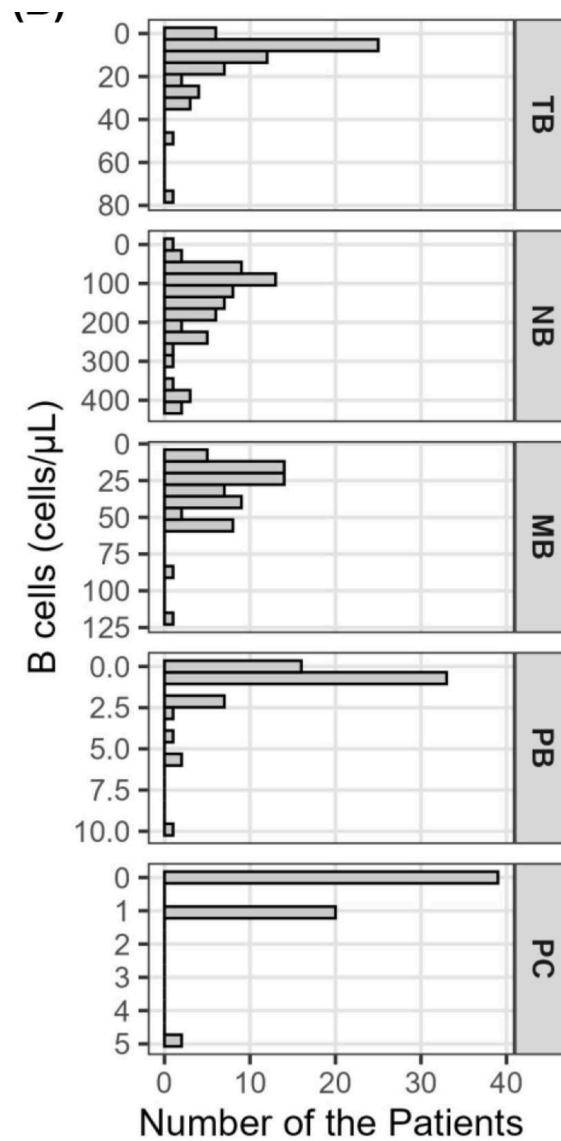
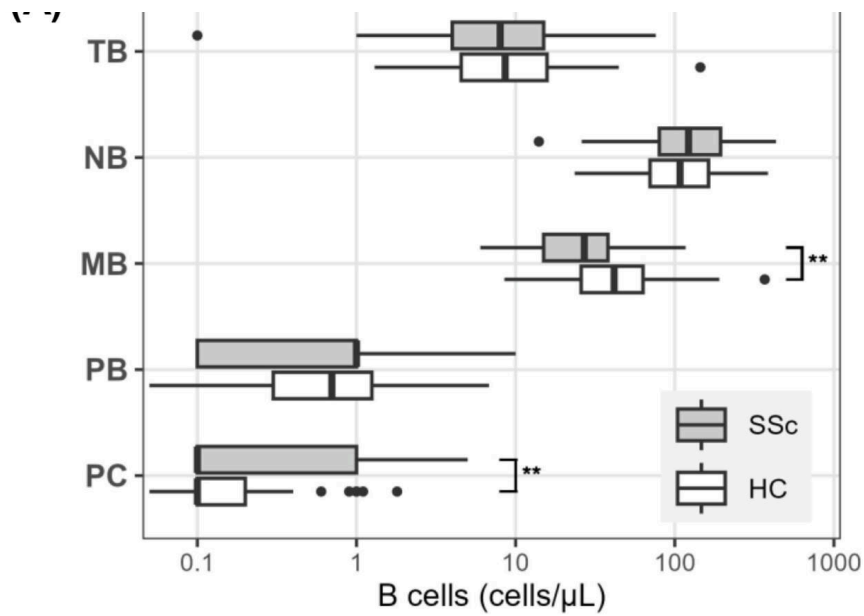


Figure 3: Distribution of B cell subsets in RWD from patients with SSc. (A) Comparison of B cell subsets between SSc patients (n= 61) and healthy controls (HC) (n= 51).

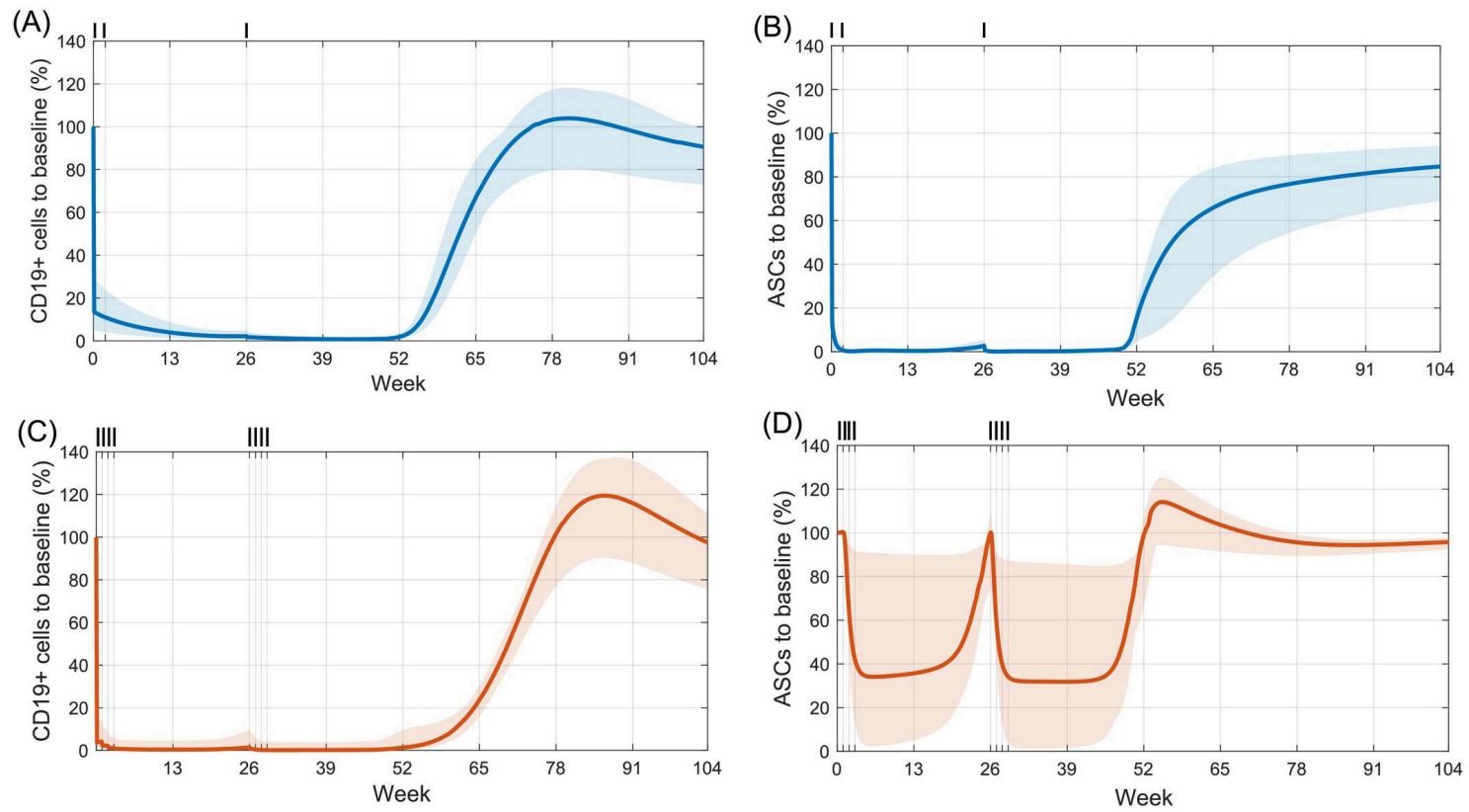


Figure 4: Simulated time courses of CD19+cells and ASCs in patients with SSc administered INEB and RTX. Lines and shades represent median and range of CD19+cells (A, C) an

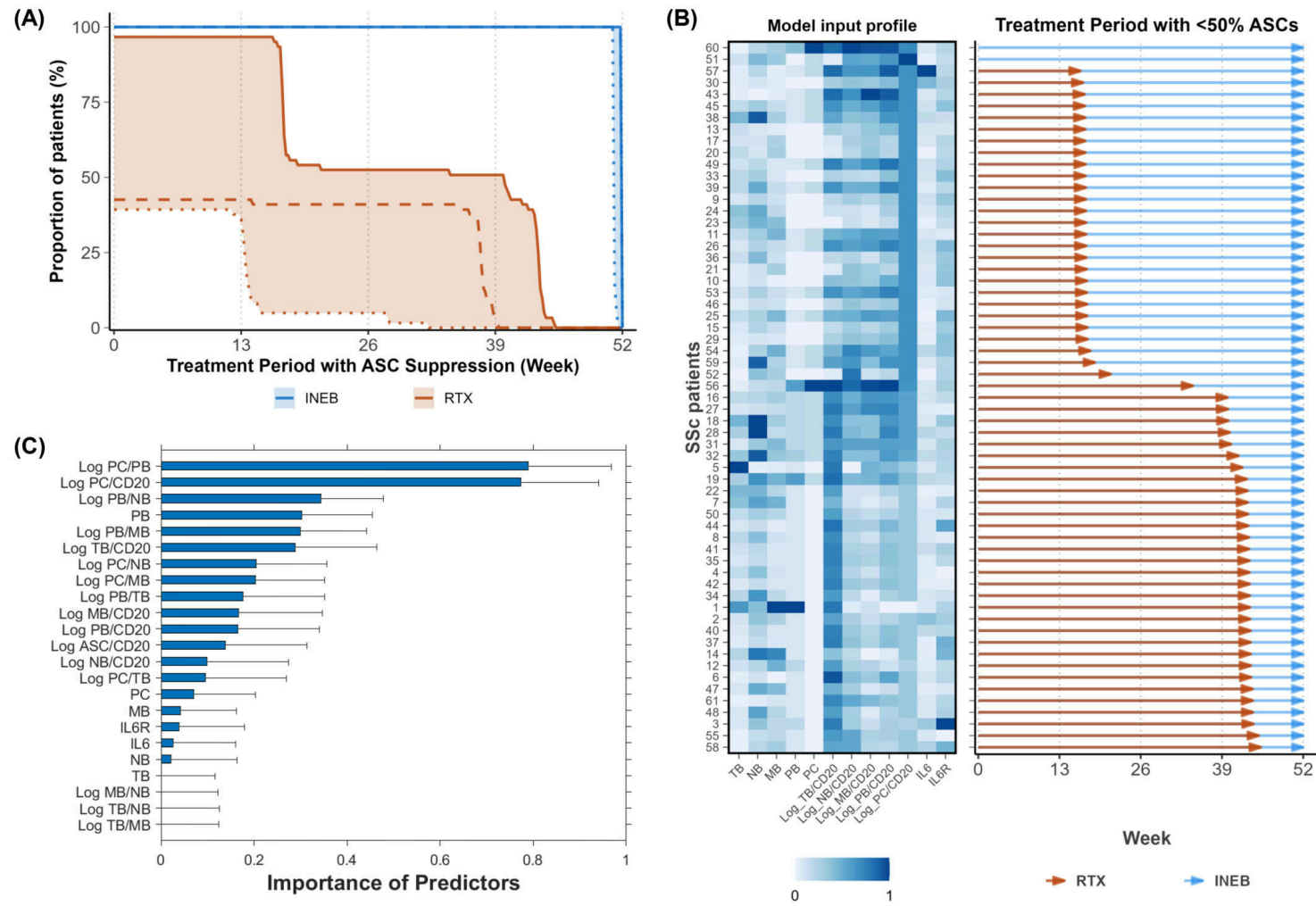


Figure 5: Simulated suppression of ASCs and key predictors to stratify ASC response to each drug. (A) Proportion of the virtual patients (VPs, n= 61) achieving $\geq 50\%$ (solid